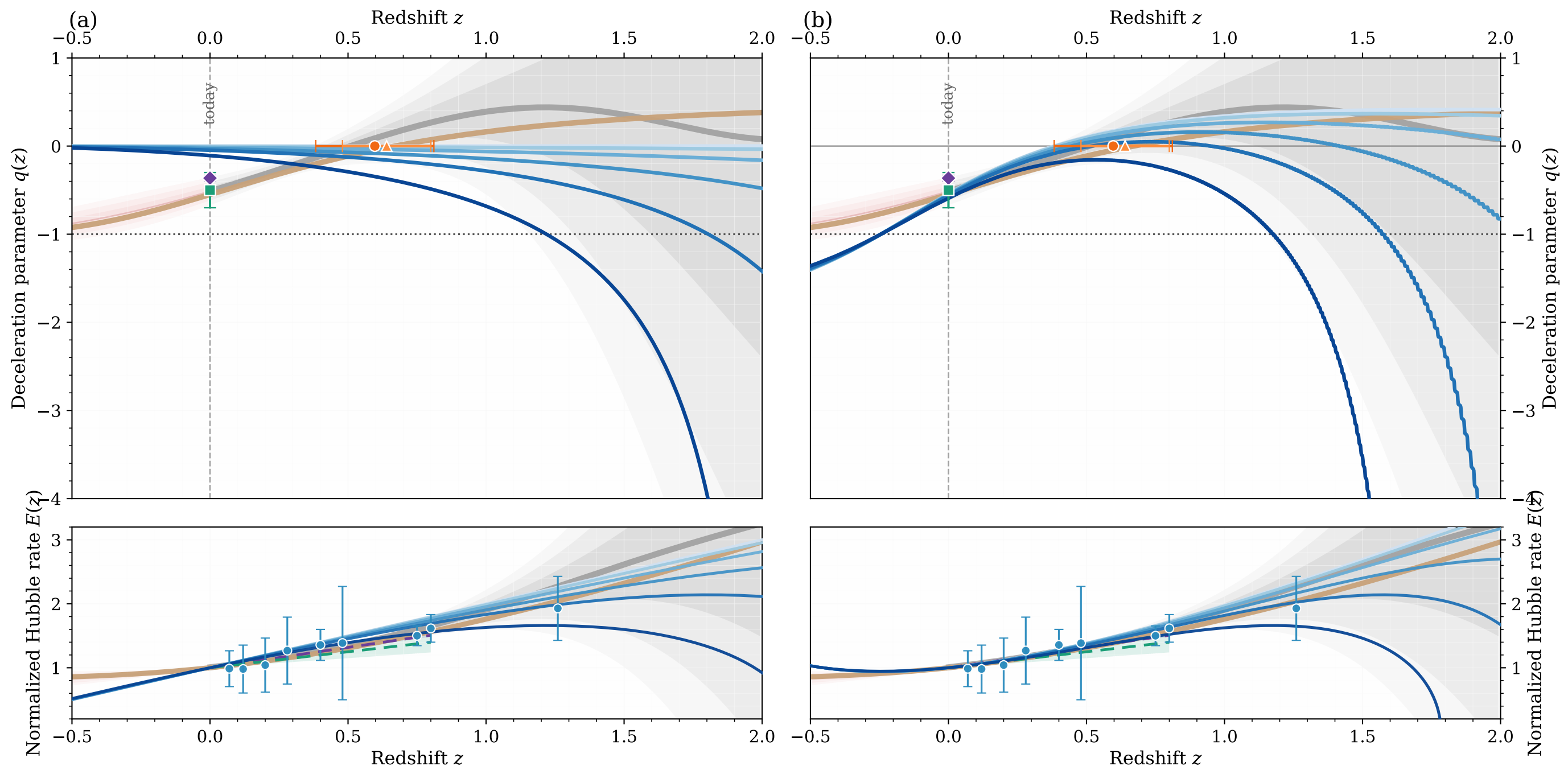




Models

- Hippopede: $\chi = 0^\circ$ & hyperconical
- Hippopede: $\chi = 15^\circ$
- Hippopede: $\chi = 30^\circ$
- Hippopede: $\chi = 45^\circ$
- Hippopede: $\chi = 60^\circ$
- Hippopede: $\chi = 85^\circ$
- Standard model: flat Λ CDM + radiation
- de Sitter / inflation: $q = -1$

Data

- Cosmic chronometers $H(z)$ data (Table 1)
- ◆ Observed q_0 (CC+Pantheon+SH0ES+BAO; Myrzakulov et al. 2025)
- Observed q_0 (FRB+SNe cosmography; Gao et al. 2024)
- Observed z_t (CC+Pantheon+SH0ES+BAO; Myrzakulov et al. 2025)
- ▲ Observed z_t ($H(z)$ meta-analysis; Hu et al. 2025)
- - Tangent from q_0 (Myrzakulov et al. 2025)
- - Tangent from q_0 (Gao et al. 2024)

GaPP CC+Pantheon+

- GaPP median
- GaPP 1σ
- GaPP 2σ
- GaPP 3σ
- GaPP extrapolation